

Likelihood-Ranked Selection in Self-Consuming Diffusion: A W_2 Negative Result and a Narrow Minority-Retention Regime

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Abstract

Self-consuming training loops can induce *model collapse*. We study unlabeled likelihood ranking of synthetics in self-consuming DDPM training on imbalanced mixtures (workshop-scale; not ImageNet diffusion). **Primary:** top- k does *not* improve sliced W_2 vs. random- k at generation $g=1$ ($\Delta W_2 > 0$ on 38/38 cells; bootstrap 95% CI excludes zero). **Secondary:** after falsifying an earlier “false stability” hypothesis, minority retention r shows bottom>rand>top mainly at moderate imbalance $\rho=5$ (exploratory $n=10$, 10/10, exact $p=0.002$; held-out seeds 10–14: 4/5, CIs still exclude zero for both contrasts). Full order is weaker at $\rho=7$ (6/10) and $\rho=10$ (4/13). Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an interventional selection policy at matched synthetic count and matched real-mix α .

Caveats that apply throughout: this is a workshop empirical note on controllable mixtures; the retention ordering is post-hoc relative to an earlier falsified hypothesis (Section 6); we do not claim ImageNet-scale generality.

Contributions. (1) Primary powered W_2 negative result for top- k at $g=1$ ($n=38$ cells). (2) Stratified retention at $\rho \in \{5, 7, 10\}$, with a labeled held-out seed block at $\rho=5$. (3) A continuous ρ sweep and a lightweight account of why the full order weakens as ρ grows (minority-mass floor). (4) Accumulate, exact-density oracle, and learned real-only verifier baselines at $\rho=5$, plus GMM precision/recall and a CIFAR FID check; explicit contrast to siloed biased-verifier selection [Qiao et al., 2026].

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

Table 1: Differentiation vs. Qiao et al. [2026] (sample-selection bias precipitating collapse).

	Qiao et al. [2026]	This note
Verifier	Biased <i>local real</i> reference (silos)	Generator’s <i>own</i> likelihood proxy
Setting	Multi-party / privacy silos	Single self-consuming loop
Intervention	Local vs. collaborative Wasserstein proxy	Matched top / rand / bottom- k
Primary diversity metric	Variance / diversity decay	Mode masses r + sliced W_2
Main message	Local filtering can accelerate collapse	Top- k harms W_2 ; r -order is ρ -dependent

3 Method

3.1 Notation

Table 2: Symbols used throughout.

Symbol	Meaning
ρ	Train imbalance: majority mode weight boost before renormalization
g	Self-consuming generation index ($g=0$ is the first model on real data)
r	Minority-mass retention $m^{(1)}/m^{(0)}$ at consecutive generations
α	Fraction of each train set drawn from a fixed real buffer
k_{frac}	Selection-pool size as a fraction of N synthetics
$s(x)$	Denoising-MSE likelihood proxy (lower = higher likelihood)

3.2 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.3 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}}N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.4 Metrics

Nearest-mean mode masses; minority-mean m ; retention $r = m^{(1)}/m^{(0)}$ (allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests: bootstrap 95% CIs of the mean paired Δ ; Wilcoxon signed-rank with exact sign enumeration for $n \leq 16$ and a normal approximation for larger n . Cells with sliced $W_2 > 10$ (pathological late-generation collapse) are excluded from W_2 tables only.

Table 3: $\Delta W_2(\text{top} - \text{rand})$ by generation (positive = top worse). Bootstrap 95% CI; Wilcoxon; Holm across generations. One $g=5$ cell excluded ($W_2 > 10$).

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	p (Holm)
1	+0.087	0.075	1.16	[0.064, 0.112]	38/38	$< 10^{-6}$
2	+0.043	0.044	0.96	[0.029, 0.057]	32/38	5×10^{-6} (1.5×10^{-5})
3	+0.038	0.039	0.98	[0.026, 0.051]	36/38	1×10^{-6} (3×10^{-6})
4	+0.022	0.044	0.49	[0.008, 0.035]	29/38	0.00029
5	+0.027	0.030	0.91	[0.018, 0.037]	31/37	6×10^{-6} (1.5×10^{-5})

Table 4: Per- ρ mean r and full-order counts (generation $g=1$).

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	15	0.604	0.350	0.128	14/15	15/15 / 14/15
7	10	0.372	0.248	0.113	6/10	9/10 / 8/10
10	13	0.248	0.183	0.120	4/13	10/13 / 8/13

4 Experiments

4.1 Design ($n=38$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Log: `w2_fs.jsonl`. Seeds: $\rho=5$ uses 0–14 (0–9 exploratory; 10–14 held-out); $\rho=7$ uses 0–9; $\rho=10$ uses 0–12.

4.2 Sliced W_2 (primary)

Pooled cells share ρ strata and are not i.i.d. Bernoulli trials; primary inference is the paired Δ CI / Wilcoxon over matched (seed, ρ) units, not a binomial over cells.

4.3 Retention (secondary; stratified)

Held-out seeds 10–14 were not used during exploratory search; they replicate the *direction* (CIs exclude zero; full order 4/5) but, at $n=5$, exact Wilcoxon remains floored for bottom–rand. At higher ρ , top still tends to underperform random, but the full three-way order is no longer reliable.

Bottom- k quality cost. Mean sliced W_2 at $g=1$ ($\rho=5$, $n=15$): bottom 0.51 < rand 0.64 < top 0.80. Bottom- k improves aggregate W_2 while retaining minorities—it is not a “junk dump” on this metric (same ordering at $\rho=7, 10$).

Continuous ρ sweep (seeds 0–4). Full bottom>rand>top rate: 5/5 at $\rho \in \{3, 4, 5\}$, then 3/5 ($\rho=6$), 2/5 ($\rho=7, 8$), 1/5 ($\rho=10$) (Figure 2).

Baselines at $\rho=5$ (seeds 0–4). Growing accumulate [Gerstgrasser et al., 2024] (cap 3000): mean $r=0.41$, $W_2=0.65$ —better than replace, but below bottom- k on both metrics. Exact-density oracle top/bottom (GMM analogue of an external likelihood verifier): $r=0.17/0.71$, nearly matching proxy top/bottom (0.18/0.74), so the selection effect is not an artifact of a broken proxy. Learned exterior

Table 5: Paired retention contrasts vs. random- k (bootstrap 95% CI; Wilcoxon).

Stratum	Contrast	Mean Δ	d_z	95% CI	p
$\rho=5$ (all)	top-rand	-0.222	-1.76	$[-0.28, -0.16]$	0.0001
$\rho=5$ (all)	bottom-rand	+0.254	+1.96	$[0.19, 0.32]$	0.0001
$\rho=5$ expl. 0-9	top-rand	-0.225	-1.82	$[-0.29, -0.15]$	0.002
$\rho=5$ expl. 0-9	bottom-rand	+0.285	+2.08	$[0.21, 0.37]$	0.002
$\rho=5$ held-out 10-14	top-rand	-0.216	-1.49	$[-0.32, -0.09]$	0.125
$\rho=5$ held-out 10-14	bottom-rand	+0.191	+2.01	$[0.11, 0.25]$	0.0625
$\rho=7$	top-rand	-0.135	-1.58	$[-0.19, -0.08]$	0.0039
$\rho=7$	bottom-rand	+0.124	+0.88	$[0.05, 0.21]$	0.0195
$\rho=10$	top-rand	-0.063	-0.71	$[-0.11, -0.02]$	0.037
$\rho=10$	bottom-rand	+0.065	+0.44	$[-0.00, 0.15]$	0.204

verifier (DDPM fit only on the fixed real buffer; Feng/Cai-style external score): $r=0.17/0.74$, again matching generator-proxy ranking. Unranked mix: $r=0.37$, $W_2=0.69$.

Precision / recall (GMM, $k=5$). At $g=1$, top- k has the highest improved precision (0.96) but worst W_2 and retention; bottom- k trades a little precision (0.88) for better W_2 (0.49) and minority r (0.74). Recall stays high for all policies (≈ 0.98).

Proxy vs. exact GMM density. On $g=0$ synthetics ($\rho=5$, $n=5$), Spearman of $s(x)$ vs. $-\log p_{\text{GMM}}(x)$ averages 0.44 (range 0.33-0.51); top- k Jaccard vs. exact density ranking averages 0.64 at $k=N/2$. The proxy is imperfect pointwise, yet oracle/verifier-vs-proxy *policies* agree closely (above).

Illustrative α / k_{frac} ablations ($n=2$) and 8D GMM / Digits-PCA transfers ($n=3$) are in Appendix A. Digits pixel-DDPM, Digits-CVAE, and CIFAR-10 tiny-UNet are reported there as preliminary/negative, not as main transfer claims. CIFAR seed-0 Inception FID at $g=1$: top 477, rand 489, bottom 332 (same $g=0$ baseline 329)—bottom- k preserves FID while top/rand collapse.

5 Analysis: why the order weakens with ρ

Top- k keeps low- $s(x)$ synthetics, which under an imbalanced teacher are disproportionately majority-mode. As ρ grows, minority mass at $g=0$ already shrinks, so the ranked pool contains few minority candidates to prefer or discard: top/random/bottom selection become more similar, r contrasts shrink, and the full bottom>rand>top pattern becomes seed-noisy—a minority-mass floor, not a contradiction of the $\rho=5$ regime. Aggregate W_2 remains a poor minority monitor: top- k is worse on W_2 at $g=1$ across the whole matrix even where the three-way r -order fails.

6 Limitations

Scale. 2D/8D GMMs only for primary claims; CIFAR tiny-UNet / FID is appendix-only ($n=1$ seed for FID); no ImageNet / CIFAR-10-LT / DiT result. **Process.** Documented exploratory search (>100 killed directions in docs/KILL_LOG.md) precedes this topic; within-topic, the original false-stability hypothesis was falsified and the retention claim revised (HARKing risk). Mitigation: W_2 primary; stratified ρ ; held-out seeds 10-14 at $\rho=5$; bootstrap CIs; continuous ρ sweep on seeds 0-4. **Other.** α/k_{frac} ablations are $n=2$; accumulate/oracle/verifier/PR baselines are $n=5$ at $\rho=5$;

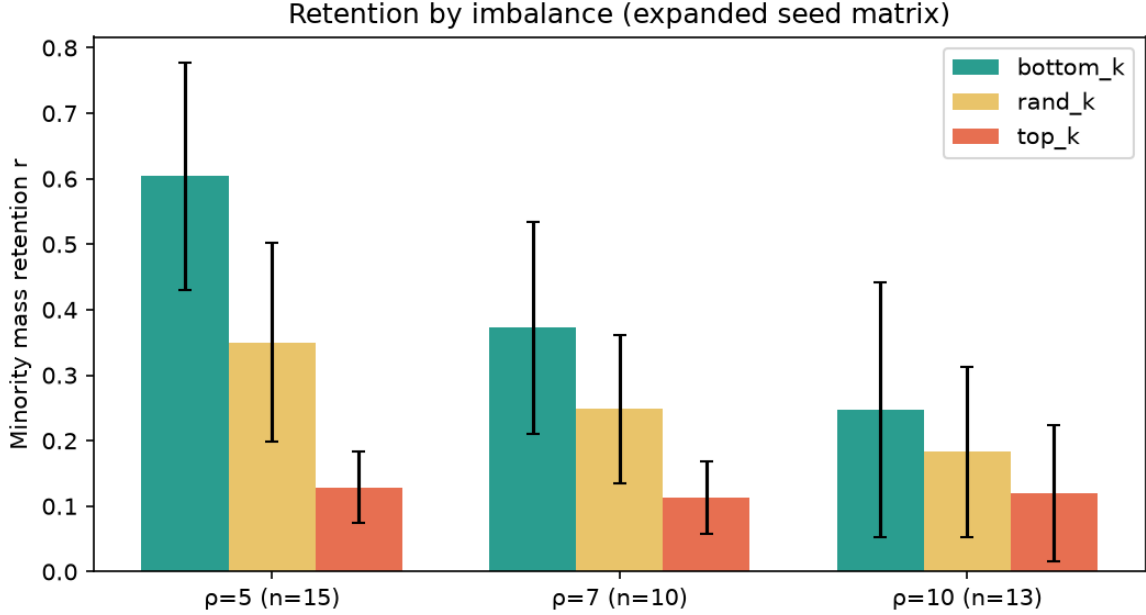


Figure 1: Retention r by ρ (means $\pm$ sd over seeds; primary matrix).

learned verifier is a real-only DDPM (not a Cai-style latent probe); Holm within W_2 generations and within paired $\rho=5$ contrasts.

7 Conclusion

Top- k likelihood selection does not improve sliced W_2 at $g=1$ in this matrix. Minority-retention ordering is strongest at moderate imbalance and weakens as minority mass floors—a mechanism-level boundary, not a universal law.

References

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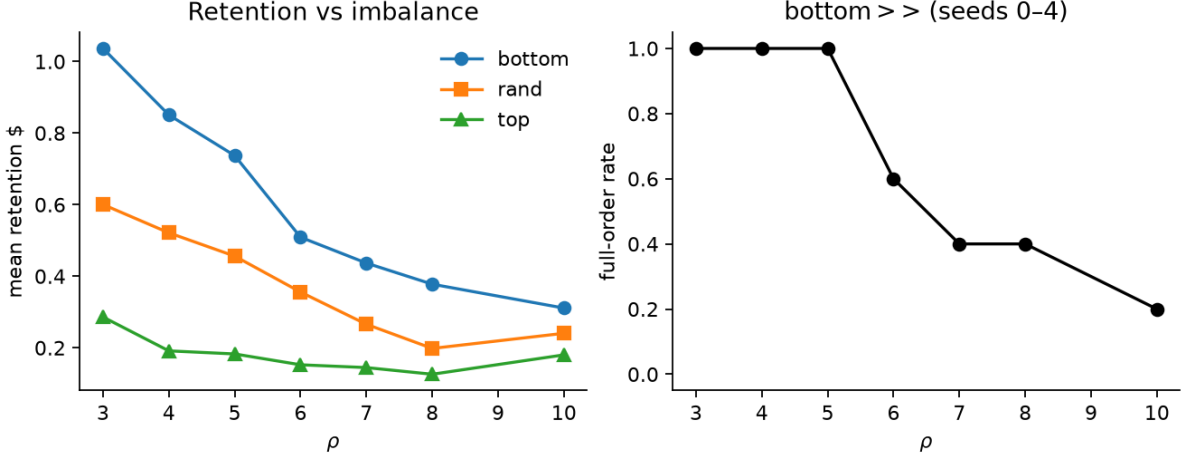


Figure 2: Continuous ρ sweep (seeds 0–4): mean r (left) and full-order rate (right).

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A Extra ablations and preliminary image checks

α and k_{frac} ($n=2$, illustrative). At $\rho=5$, order holds 1/2 at $\alpha \in \{0.25, 0.50\}$ and 0/2 at $\alpha=0.75$; $k_{\text{frac}}=0.25$ coincides with 0.5 when n_{syn} binds, and 0.75 breaks full order (0/2). Default rows of the two ablations are the same experimental cell.

Directional transfers ($n=3$). 8D GMM and Digits-PCA: full order 3/3 (exact p floor 0.25). **Preliminary / negative:** Digits pixel-space DDPM full order 1/3; Digits-CVAE inconclusive. CIFAR-10 4-class imbalanced tiny-UNet DDPM ($\rho=5$, $n=3$ seeds, CPU, 512 images, 400 train steps): mean retention $\bar{r}$ is bottom 1.53 > rand 1.38 > top 0.93, but seed-wise full order 0/3 (nearest class-mean masses; mode assignment is noisy on blurry samples). Seed-0 Inception FID at $g=1$: bottom 332 vs. top 477 / rand 489 (shared $g=0$ FID 329). These do not support a main-text transfer claim.

B Reproducibility

python scripts/reproduce_w2.py; python experiments/scripts/make_submission_stats.py.
Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.